

Chujun Zong

Dr.-Ing. · Life Cycle Assessment · Sustainable Building Design

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RESEARCH PROFILE

Doctoral researcher at the Technical University of Munich specializing in dynamic Life Cycle Assessment (LCA) methodology for the built environment. Author of 5+ first-author articles in JCR Q1 journals (Automation in Construction, Building and Environment, Scientific Reports), with research bridging multi-objective stochastic optimization, uncertainty modeling, and graph-database-driven dynamic LCA (GraphDLCA). Trilingual in English, German, and Mandarin.

RESEARCH INTERESTS

Dynamic life cycle assessment · Multi-objective stochastic optimization under uncertainty · Graph-database modeling for building inventories · Early-stage design decision support · EU sustainability regulation.

EDUCATION

Dr.-Ing. Technical University of Munich (TUM) <i>Chair of Energy Efficient and Sustainable Design and Building.</i> <i>Supervisor: Prof. Werner Lang. Passed with Magna Cum Laude</i>	2022 – 2026
M.Sc. Resource Efficient and Sustainable Building, Technical University of Munich (TUM)	2019 – 2021
B.A. Architecture, Technical University of Munich (TUM)	2014 – 2019
Exchange B.A. Architecture, École Polytechnique Fédérale de Lausanne (EPFL)	2016 – 2017

PROFESSIONAL EXPERIENCE

Research Associate & Doctoral Candidate — Technical University of Munich <i>Chair of Energy Efficient and Sustainable Design and Building (Prof. Werner Lang, TUM Vice President for Sustainable Transformation). Lead two BMWK/DFG-funded research projects; supervise master's theses; teach across English and German curricula.</i>	2022 – 2026
Working Student — Siemens Energy AG, Construction, Design & Sustainability <i>Developed internal CO₂ pricing methodology and contributed to corporate sustainability and workplace guidelines.</i>	2020 – 2021

FUNDED RESEARCH PROJECTS

CircularGreenSimCity <i>Holistic life cycle assessment of urban districts. Funded by the German Federal Ministry for Economic Affairs and Climate Action (BMWK).</i>	2022 – 2025
EarlyBIM2 <i>Knowledge database and machine learning assistance for performance-oriented building. Funded by the German Research Foundation (DFG).</i>	2022– 2024

TEACHING EXPERIENCE

System Effects and Interdependencies of Sustainable Planning in Civil Engineering (English) <i>Teaching Assistant, Technical University of Munich</i>	2025 – 2026
Introduction to Scientific Writing (English) <i>Lecturer, Technical University of Munich</i>	2023 – 2026
Sustainable Building & Ecological Building — Basic Modules (German) <i>Teaching Assistant, Technical University of Munich</i>	2022 – 2026
Sustainable Lighting Technology (English) <i>Lecturer, Technical University of Munich</i>	2022 – 2025

HONORS & AWARDS

- Doce et Delecta Teaching Award, First Place — Faculty of Civil and Environmental Engineering, TUM (Winter Semester 2022/2023, Sustainable Lighting Technology)
- National Award for Outstanding Self-funded Students Abroad (Category A), 2024 – China's highest honor for overseas doctoral researchers

ACADEMIC SERVICE

Peer reviewer for international journals including Building and Environment, Building Simulation, Buildings, Sustainability, Journal of Architectural Engineering and Design Management, Scientific Reports, Discover Artificial Intelligence, Cell Reports Sustainability, etc.

SELECTED PUBLICATIONS

Peer-Reviewed Journal Articles

- **Zong, C.**, Banihashemi, F., Petzold, F. & Lang, W. (2026). Graph-based Database Structure and Calculation Framework for Automated Dynamic Life Cycle Assessment of Buildings. Accepted for publication in *Automation and Construction*. (JCR Q1)
- **Zong, C.**, Banihashemi, F., & Lang, W. (2025). Dynamic life cycle impact assessment (DLCIA) in a sustainable building planning process. *Scientific Reports*, 15(1), 32680. (JCR Q1)
- **Zong, C.**, Chen, X., Deghim, F., Staudt, J., Geyer, P., & Lang, W. (2024). A holistic two-stage decision-making methodology for passive and active building design strategies under uncertainty. *Building and Environment*, 251. (JCR Q1)
- **Zong, C.**, Margesin, M., Staudt, J., Deghim, F., & Lang, W. (2022). Decision-making under uncertainty in the early phase of building façade design based on multi-objective stochastic optimization. *Building and Environment*, 226, 109729. (JCR Q1)
- Napps, D.; Staudt, J.; Saluz, U.; Chen, X.; Steiner, D.; Zong, C.; Deghim, F.; Lang, W.; Geyer, P.; Schnellenbach-Held, M.; Petzold, F.; Borrmann, A.; König, M. (2026). Evaluation of Building Design Variants in Early Phases on the Basis of Adaptive Detailing Strategies. *Buildings*, 16(4), 685. (JCR Q2)
- Banihashemi, F., Weber, M., Deghim, F., **Zong, C.**, & Lang, W. (2024). Occupancy modeling on non-intrusive indoor environmental data through machine learning. *Building and Environment*, 254. (JCR Q1)

Selected Conference Papers

- **Zong, C.**, & Lang, W. (2025). Dynamic factors in life cycle assessment: systematic review and categorization. Building Simulation Conference Proceedings 2025.
- **Zong, C.**, Lang, W., Pancheri, F. & Sun, Y. (2025). Assessing pedestrian level barriers using mobile robot to validate the 15-minute city concept. Building Simulation Conference Proceedings 2025.
- **Zong, C.**, Sun, Y., & Lang, W. (2024). System Dynamics Modeling of Life Cycle Carbon Footprints for Building Wall Insulation Materials. IOP Conference Series: Earth and Environmental Science, 1363.
- **Zong, C.**, Reitberger, R., Deghim, F., Staudt, J., Larkin, J. A., & Lang, W. (2023). A comparative study of dynamic building simulation and machine-learning within a two-stage multi-objective stochastic optimization framework. Building Simulation Conference Proceedings, 18, 1540–1547.
- **Zong, C.**, Banihashemi, F., Vollmer, M., & Lang, W. (2022). Implementation of occupant behaviour models for window control using co-simulation approach. BauSIM 2022.